longer patent their drugs in India. What was allowed were process patents—the process of manufacturing a drug using active pharma ingredients—which became a big turning point, enabling domestic firms to make the same products as the multinationals, but through a reverse engineering process. This meant that drugs thus made did not infringe on the patents of their branded counterparts, but at the same time, had similar ingredients that made them as effective.

The '80s and the '90s saw the morphing of the Indian pharma industry into a global giant. Companies like Cipla, Dr. Reddy's Laboratories, Ranbaxy Laboratories and Sun Pharma led the generic drug revolution, exploring the overseas markets and challenging the branded drugs of MNCs that were nearing patent expiry in the US market. The successful challengers, termed 'generic drugs' or 'generics', cost a fraction of what a brand-name drug would, and now comprise over 80 per cent of all prescriptions in the US. There were breakthrough generics from India that created a sensation in the pharma world too. For instance, Cipla developed the world's first 3-in-1 fixed dose combination of anti-AIDS drugs in 2001, and offered to supply this cocktail to developing countries at a fraction of the cost in the West. This led to a big revolution in AIDS treatment.

Gradually, the time gap between the development of drugs born out of original innovation by MNCs and their generic substitutes from India shrunk. India joined the Trade-Related Aspects of Intellectual Property Rights (TRIPS) Agreement on January 1, 1995. The agreement required India to acknowledge product patents. Domestic firms took this as an opportunity to boost R&D and search for their own innovative drugs, a process that continues to this day.

In the biotech arena, Biocon has put India on the world map for biopharmaceuticals, developing the first biosimilar (a biotech product similar to another already approved) recombinant human insulin in 2004 and India's first indigenously produced novel monoclonal antibody (a man-made protein) for head and neck cancer. Serum Institute of India shot into the limelight for manufacturing the anti-Covid vaccine Covishield, from Oxford-AstraZeneca in 2021. The Jan Aushadhi programme, a government initiative launched in 2008, to provide affordable, unbranded quality medicines to poor people in India through government-supported pharmacies, has furthered the use of generic medicines in India. ■

DOSE OF DOMINANCE

» **India supplies nearly 40% of the generic medicines sold in the US, the world's largest pharma market. It also accounts for one-fourth of the generic medicine supplies in the UK**

» **India leads globally with the highest number of USFDA-compliant pharma manufacturing facilities, totalling 670 in 2023**

» **It is also home to 500 active pharmaceutical ingredient (API) producers, contributing about 8% to the global API market**

» **As much as 70% of India's demand for medicines is met by domestic companies**

» **India supplies over 50% of the global demand for vaccines**

» **80% of the antiretroviral drugs used globally to combat AIDS are supplied by Indian firms**

$50 BN
(Rs 4.2 lakh crore)
Size of the Indian pharma industry, the world's third largest. It manufactures about 60,000 different generic brands across 60 therapeutic categories and exports to over 200 countries



FROM THE ARCHIVES
Jun. 30, 1976

Cover story
Drug Industry in the Doldrums

The magnitude of India's population offers vast opportunities and challenges to the drug industry. However, these challenges are common to both manufacturers of medicines and administrators of health services. In the ultimate analysis, the success of a drug policy will depend on the social purpose and dynamism of all concerned—the government, the industry and the medical profession. Medicine is a matter of ethics and trust. Trust may yet beget trust. Ethics presupposes discipline.
–W.S. Titus